

Efe Onaran

CONTACT INFORMATION	David Rittenhouse Laboratory, Office: 4N55 209 S 33rd St, Philadelphia, PA 19104	eonaran.github.io eonaran@seas.upenn.edu
RESEARCH INTERESTS	Probability theory with a focus on the topology and geometry of random high-dimensional objects, and random growth models.	
ACADEMIC APPOINTMENTS	University of Pennsylvania • Visiting Scholar/Postdoctoral Research Fellow Joint appointment at the Department of Mathematics and the Department of Electrical and Systems Engineering • Mentors: Robin Pemantle, Yuliy Baryshnikov, Daniel E. Koditschek	2024–present
	University of Illinois Urbana-Champaign • Postdoctoral Research Fellow at the Coordinated Science Laboratory • Mentor: Yuliy Baryshnikov	2023–2024
	Technion – Israel Institute of Technology • Postdoctoral Research Fellow • Mentors: Omer Bobrowski, Robert J. Adler	2020–2023
EDUCATION	New York University • Ph.D. in Electrical Engineering • Thesis title: Recovery of Permutational Data: Statistical Limits, Efficient Algorithms and Applications • Advisor: Sundeep Rangan	2015–2020
	Princeton University • M.Sc. in Electrical Engineering • Thesis title: Stochastic Block Model with Side Information • Advisor: Emmanuel Abbe	2013–2015
	Bilkent University • B.Sc. in Electrical Engineering	2008–2013
JOURNAL PUBLICATIONS & PREPRINTS	Yuliy Baryshnikov and Efe Onaran. Ising disks: topology preserving glauher dynamics. <i>J. Appl. Comput. Topol.</i> , 10(3):Paper No. 16, 2026	
	Efe Onaran, Omer Bobrowski, and Robert J. Adler. Central limit theorems for local functionals of dynamic point processes. <i>Electronic Journal of Probability</i> , 31, 2026	
	Efe Onaran, Omer Bobrowski, and Robert J Adler. Functional central limit theorems for local statistics of spatial birth–death processes in the thermodynamic regime. <i>The Annals of Applied Probability</i> , 33(5):3958–3986, 2023	
	Efe Onaran and Soledad Villar. Shuffled linear regression through graduated convex relaxation. <i>arXiv:2209.15608</i> , 2022	
	Jinfeng Du, Efe Onaran, Dmitry Chizhik, Sivarama Venkatesan, and Reinaldo A Valenzuela. Gbps user rates using mmwave relayed backhaul with high-gain antennas. <i>IEEE Journal on Selected Areas in Communications</i> , 35:1363–1372, 2017	
	Efe Onaran, M Cengiz Onbasli, Alper Yesilyurt, Hyun Yong Yu, Ammar M Nayfeh, and Ali K Okyay. Silicon-germanium multi-quantum well photodetectors in the near infrared. <i>Optics express</i> , 20:7608–7615, 2012	

IN PREPARATION	Yuliy Baryshnikov and Efe Onaran. First-passage percolation under path constraints: efficiency–complexity tradeoffs for resource-limited agents.	
IN CONFERENCE PROCEEDINGS	Efe Onaran and Soledad Villar. Efficient belief propagation for graph matching. In <i>ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing</i> , pages 9060–9064. IEEE, 2020	
	Efe Onaran and Soledad Villar. Projected power iteration for network alignment. In <i>Wavelets and Sparsity XVII</i> , page 103941C. International Society for Optics and Photonics, 2017	
	Efe Onaran, Siddharth Garg, and Elza Erkip. Optimal de-anonymization in random graphs with community structure. In <i>2016 50th Asilomar Conference on Signals, Systems and Computers</i> , pages 709–713. IEEE, 2016	
	Efe Onaran, Marios Gatzianas, and Christina Fragouli. Broadcast erasure channel with feedback: The two multicast case—algorithms and bounds. In <i>2013 International Symposium on Network Coding (NetCod)</i> , pages 1–6. IEEE, 2013	
TALKS	AFOSR Dynamical Systems and Control Theory Review	08/2026
	Penn/Temple Probability Seminar	10/2024
	Bernoulli-IMS 11th World Congress in Probability and Statistics in Bochum	08/2024
	CRM-PIMS Montreal Summer School in Probability – Short talk	07/2024
	Koç University, Math Seminar – Online	04/2024
	University of Illinois Urbana-Champaign, Probability Seminar	10/2023
	Queen Mary University of London, Probability and Applications Seminar	12/2022
	Random Structures and Algorithms in Gniezno, Poland	08/2022
	Technion, Probability Seminar	04/2022
	Purdue University, Probability Seminar	03/2022
	International Conference on Acoustics, Speech, and Signal Processing – Online	05/2020
	Asilomar Conference on Signals, Systems and Computers	09/2016
	Sarnoff Symposium – Poster session	08/2016
TEACHING	University of Pennsylvania	
	• Instructor for Mathematics of Change - Part II (MATH 1080)	Fall 2025
	• Co-instructor for Advanced Applied Probability (MATH 5460)	Spring 2025
	• Co-instructor for Mathematics of Change - Part II (MATH 1080)	Fall 2024
	University of Illinois Urbana-Champaign	
	• Instructor for Optimum Control Theory (ECE 553 - MATH 595)	Spring 2024
	New York University	
	• Head TA for Applied Statistics (Undergraduate)	Spring 2020
	• Head TA for Probability Theory (Undergraduate)	Fall 2019
	• Head TA for Probability and Stochastic Processes (Graduate)	Spring 2016
	• Head TA for Signal Processing (Undergraduate)	Fall 2015
	• Grader for classes: Machine Learning, Optimization and Computational Linear Algebra, Inference and Representation, Mathematical Tools for Data Science	
SERVICE AND OUTREACH	Reviewer for	
	• Annals of Applied Probability, Electronic Journal of Probability, Statistics and Probability Letters, Linear Algebra and its Applications, IEEE/ACM Transactions on Networking, Discrete Applied Mathematics, International Symposium on Information Theory	
	Tutoring	

- High school math tutoring for Regents Exams, Opportunity Charter School, NY 2018

HONORS AND AWARDS	Weber Fellowship at NYU Tandon School of Engineering	2015-2016
	Princeton University Department of Electrical Engineering Graduate Fellowship	2013-2014
	Highest CGPA (3.97/4.00) among all 2013 graduates of Bilkent U., School of Eng.	2013
	Full scholarship at Bilkent University	2008-2013
	Prime Ministry of Turkey Scholarship	2008-2013
CITIZENSHIP	Türkiye, United States	